



Amazon Data Firehose

Streaming Data Made Simple

AWS Machine Learning Associate • Part 4

Presented by Prajan Shrestha

What We'll Cover



What is Amazon Data Firehose?



How it works, step by step



Firehose vs. Kinesis Data Streams



Where the data comes from



Where the data goes



Key features & common use cases

What is Amazon Data Firehose?

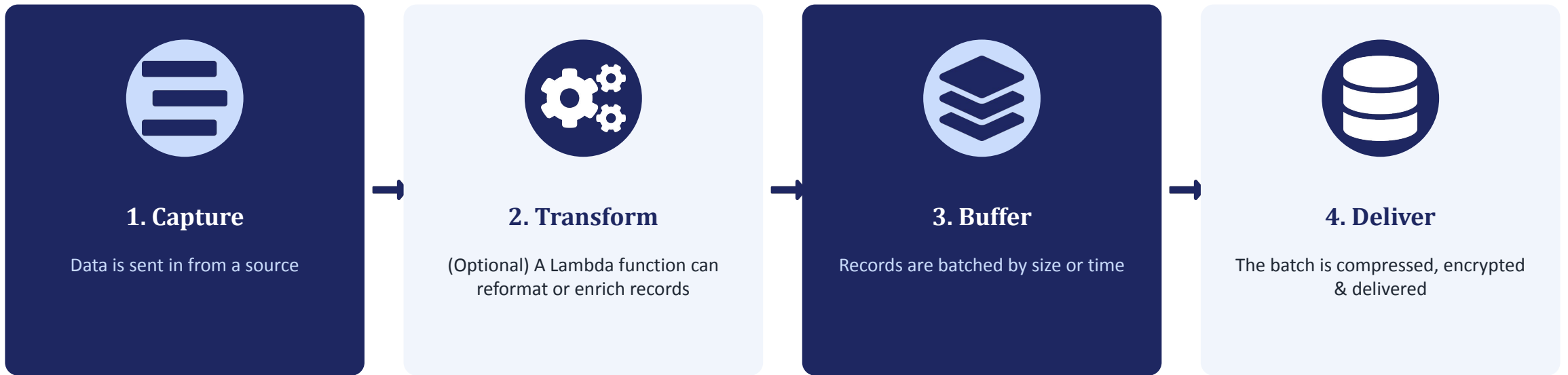


Formerly "Kinesis Data Firehose" — same service, new name.

A fully managed service for delivering real-time streaming data to a destination of your choice.

It can capture data, optionally transform it along the way, and load it automatically, all without you writing an application or managing any servers.

How It Works



Buffering is based on whichever threshold is hit first: buffer size or buffer time.

Firehose vs. Kinesis Data Streams



Data Firehose

Fully managed delivery. You just pick a destination, no consumer code to write.



Kinesis Data Streams

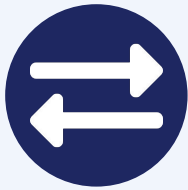
You manage the consumers and processing logic yourself, for full control.

Where the Data Comes From



Kinesis Data Streams

Already streaming data through
Kinesis



Direct PUT

Apps send records via SDK or API



CloudWatch Logs & Events

Operational & monitoring data



AWS IoT

Data from connected devices

Where the Data Goes



Amazon S3

Storage & data lakes



Amazon Redshift

Data warehousing



Amazon OpenSearch

Search & log analytics



Splunk & HTTP endpoints

Third-party tools

Key Features



Fully managed & serverless

No infrastructure to provision or patch



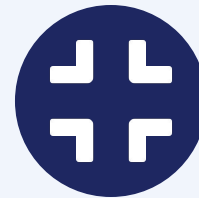
Auto-scaling

Matches whatever throughput you throw at it



Near real-time delivery

Typically within seconds



Built-in processing

Compression, encryption & format conversion (e.g. to Parquet)

Common Use Cases



Streaming ETL

Continuously loading data into a data lake as it arrives



Log & Event Analytics

Capturing application or system logs for analysis



IoT Data Ingestion

Handling high-volume sensor data



Real-Time Dashboards & ML

Feeding pipelines that need fresh data instantly

Recap

- A fully managed way to capture, transform & deliver streaming data.
- Sources like Kinesis, IoT & CloudWatch flow in automatically.
- Destinations like S3, Redshift & OpenSearch, no servers to manage.

Thanks for watching!